Report 6: Convolutional Neural Network

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May 23, 2026

1 Network Architecture

A convolutional neural network (CNN) was implemented from scratch using NumPy and OpenCV.

The network architecture contains:

- Convolution layer
- ReLU activation
- Max pooling layer
- Fully connected layer
- Softmax output layer

Architecture:

```
Input 28x28 -> Convolution 3x3 -> ReLU -> MaxPooling 2x2 -> Flatten
-> Fully Connected -> Softmax
```

The convolution layer extracts image features using a learnable kernel.

$$output(i, j) = \sum region \times kernel$$

The ReLU activation introduces non-linearity:

$$ReLU(x) = max(0, x)$$

Max pooling reduces the spatial size while keeping important features.
The fully connected layer performs classification.
Softmax converts outputs into probabilities:

$$Softmax(x_i) = \frac{e^{x_i}}{\sum_j e^{x_j}}$$

2 Training Pipeline

The CNN training pipeline is:

```
Image -> Convolution -> ReLU -> Pooling -> Flatten -> Fully Connected Softmax
-> Cross Entropy Loss -> Backpropagation -> Gradient Descent
```

The model was trained using gradient descent.
Weights were updated using:

$$W = W - r\nabla W$$

where:

- W is the weight matrix
- r is the learning rate
- ∇W is the gradient

Cross entropy loss was used:

$$Loss = -\log(p_y)$$

3 Dataset

The MNIST handwritten digit dataset was used.
The dataset contains 10 classes:

$$0, 1, 2, \ldots, 9$$

Each image has size:

$$28 \times 28$$

4 Evaluation Metrics

The network was evaluated using classification metrics.
Accuracy measures the percentage of correct predictions.

$$Accuracy = \frac{CorrectPredictions}{TotalPredictions}$$

Experimental result:

```
Accuracy = 0.84
```

5 Discussion

The CNN architecture successfully implemented:

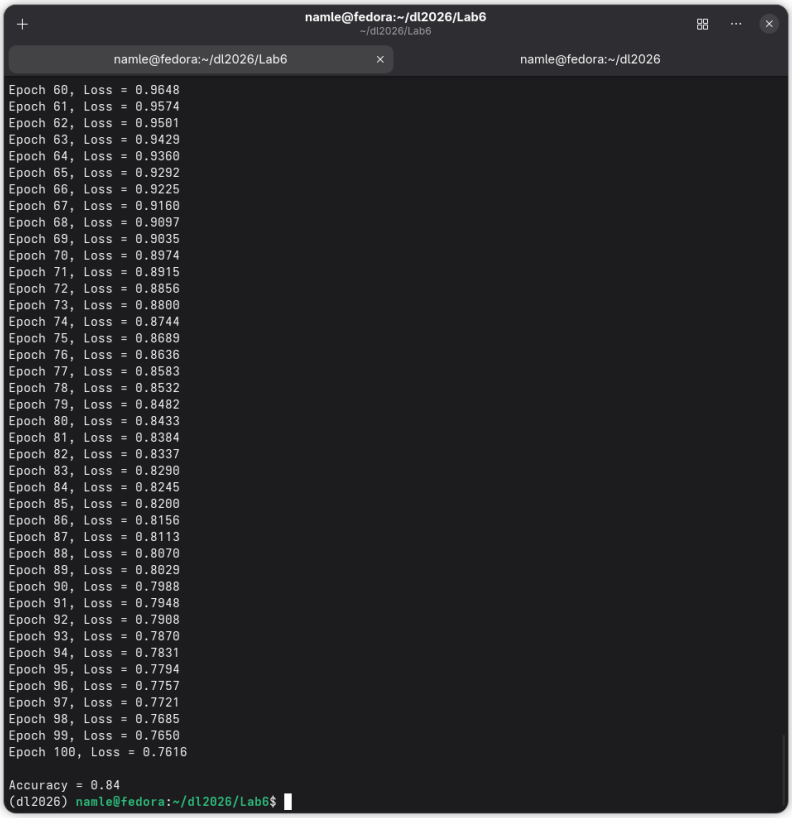
- Convolution
- Pooling
- Softmax classification
- Backpropagation
- Gradient descent

The loss decreased during training, showing that the model learned from the dataset.

The accuracy is still limited because:

- Only one convolution kernel was used
- Only the fully connected layer was trained
- The network architecture is simplified

6 Training Result Figure



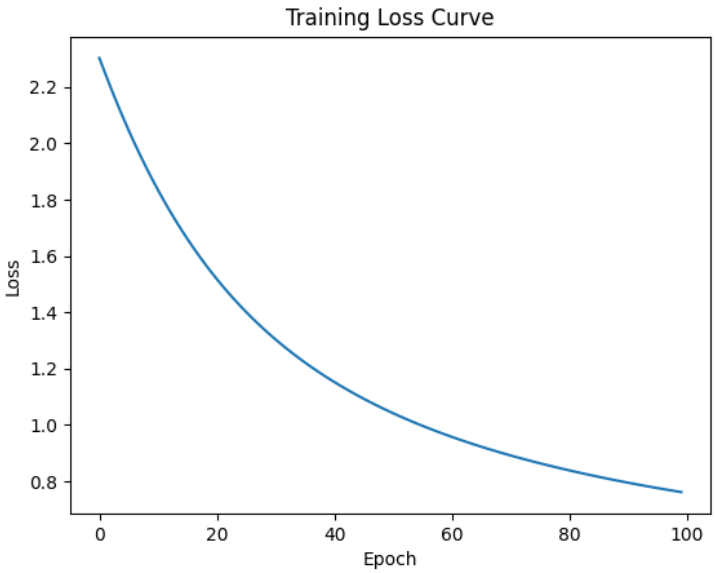


Figure 1: CNN training result and accuracy